CSC 106 - Software Engineering 1 Section
ABD1

Final Term Project



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1. Project Overview

1.1. Team & Contacts

Role	Name	Responsibilities	Contact
Project Manager	Diane Caryl C. Largoza	Oversees project timeline, scope, and stakeholder communication.	dianelargoza2@gmail.com
Technical Lead	Johnriel M. Layao	Leads system architecture, core development, and technical decisions.	layaojohnriel18@gmail.com
UI/UX Designer	Niño Carl M. Larang	Designs user interfaces, prototypes, and ensures a positive user experience.	niceshot256@gmail.com
QA Lead	Johnriel M. Layao	Develops test plans, ensures software quality and functionality.	layaojohnriel18@gmail.com

1.2. Executive Summary

XYM Mini Hardware is a small family-owned business in Butuan City that relies heavily on manual recording for sales and inventory tracking. These handwritten records create recurring problems such as miscalculations, misplaced documents, delayed stock updates, and difficulties in preparing complete inventory reports. As the store continues to serve more customers and handle larger volumes of items, the limitations of manual systems become more evident and pose risks to business accuracy and efficiency.

This project proposes the creation of a customized Inventory and Sales Monitoring System tailored to the workflow of XYM Mini Hardware. The system will automate essential processes such as daily sales computation, stock level monitoring, and low-stock, while still allowing the owner to retain manual receipts for customer service.

The system will be designed as a local-only web application that can be accessed exclusively within the store's network for security and controlled usage. By improving accuracy, reducing workload, and providing meaningful insights into stock trends, the system aims to support better decision-making and more efficient operations.

1.3. Problem Statement

Current Situation: XYM Mini Hardware uses handwritten logbooks, invoices, receipts, and withdrawal slips for recording inventory movement and sales. While this manual approach is familiar to the owner, it becomes increasingly inefficient and difficult to manage as the volume of transactions grows.

Pain Points:

- Frequent errors in the manual computation of daily sales
- Difficulty identifying stock quantities, leading to late detection of shortages
- Overstocking of slow-moving items
- Overstocking caused by canceled customer orders

- Manual workload overload for the cashier and the inventory auditor
- Limited visibility into inventory trends and movement
- Lack of automated monitoring that helps ensure accuracy in inventory records

Impact: These issues result in inaccurate sales totals, delayed restocking decisions, wasted capital due to excess inventory, and an increased workload for staff. Without automated monitoring, the store risks inefficiencies that can impact both daily operations and long-term business performance.

1.4. Project Goals & Objectives

- **Goal 1:** Improve inventory accuracy and monitoring.
  - **Objective 1.1:** Reduce inventory discrepancies by at least 70% within two months of system deployment.
  - **Objective 1.2:** Provide real-time stock level updates visible to the user within 2 seconds of any inventory adjustment.
- **Goal 2:** Automate sales computation to minimize human error.
  - **Objective 2.1:** Automatically generate daily sales totals with at least 95% accuracy compared to manual records.
  - **Objective 2.2:** Reduce cashier and inventory auditor computation workload by at least 50%.
- **Goal 3:** Provide a secure and store-only accessible system.
  - **Objective 3.1:** Restrict system access to within the hardware store's local network.
  - **Objective 3.2:** Implement role-based access control with **three** privilege levels: (Owner, cashier/inventory auditor, and Staff).

1.5. Scope & Limitations

Category	Description
In Scope	● ***Features:*** The system will include the following core functionalities: 1. Primary system workflow for monitoring inventory and daily sales 2. User authentication and role-base access control 3. Basic reporting, including daily sales summaries and inventory status 4. Storage and retrieval of inventory and sales records ● ***User Base:*** The system is intended for: 1. Primary Users: Store Owner (full access) 2. Secondary Users: Inventory Auditor/Cashier (Can process transactions & record restocks) 3. Additional User: Staff (view-only access to the dashboard and inventory lists.)

Category	Description
	● ***Data Sources:*** The system will rely on the following inputs: 1. User-submitted sales and inventory updates 2. System-generated logs or records 3. Internal computations for sales totals and inventory summaries
Out of Scope	● ***Features:*** The following functionalities are not included in this version: 1. Strictly Offline Deployment: The system is purposefully designed to be deployed offline within the XYM premises only. It is not accessible via the public internet. 2. No Mobile App (V1): The current version is a web application optimized for desktop/laptop usage in the store. 3. No External API Integration: The system does not currently sync with external supplier APIs or government tax systems automatically. 4. Official Tax Receipting: While the system now generates digital transaction slips for customer reference, official BIR receipts may still be manually issued depending on the owner's compliance preference. ***Platforms:*** The system will be developed solely as a local-only web application. Example: “A mobile application or offline mode is out of scope for Version 1.”
Constraints & Limitations	● *Technical:* 1. System performance depends on server capacity and local network stability. 2. Performance is dependent on server capacity and network connectivity. 3. Reliance on certain APIs or libraries may introduce limitations or restrictions. ● *Assumptions:* 1. Users will input accurate and complete sales/inventory data. 2. System administrators will perform routine maintenance and updates. 3. Users have basic digital literacy and access to stable internet when required.

Category	Description
	● *Known Issues:* 1. Automated or system-generated outputs may require human verification, especially for complex tasks. 2. Minor UI inconsistencies may appear and will be fixed in later updates. 3. Performance may degrade under heavy usage during peak store hours.

2. Related Review of Literature

2.1. Related Work & Technologies

This section reviews existing solutions, digital tools, and commonly used platforms that inspired or influenced the design of the proposed system. By examining their strengths and limitations, the study identifies gaps that the XYM Inventory Management System aims to address.

- **Competitive Analysis:**
  - ❖ Tool/Platform A- POS & Inventory Software (e.g., Loyverse POS)

    Strengths:

    - Provides an intuitive and beginner-friendly interface
    - Offers reliable basic automation features such as price calculations and sales logging
    - Supports essential inventory monitoring for small stores

    Weaknesses:

    - Limited customization options for niche business workflows
    - Cannot be fully tailored to very small hardware stores with unique product categories
    - Some advanced features require internet access

    Relevance:
    This tool demonstrates the value of having a simple, easy-to-use system. However, its lack of customization highlights the need for a platform built specifically for the operations of XYM Mini Hardware.

  - ❖ Tool/Platform B- Large-Scale Inventory Systems (e.g., Odoo Inventory / Zoho Inventory)

Strengths:

- Rich feature set including automation, advanced analytics, and supplier management
- Strong global community support and updated documentation
- Highly scalable for medium to large businesses

Weaknesses:

- High learning curve, especially for non-technical users
- Requires higher system resources compared to simpler solutions
- Setup and configuration can be time-consuming

Relevance:
This tool shows how powerful inventory systems can be, but it also proves that overly complex interfaces are not ideal for micro-businesses with limited staff and basic operational needs. The proposed project prioritizes simplicity without sacrificing essential functionality.

❖ Tool/Platform C- Subscription-Based Cloud Inventory Systems (e.g., QuickBooks Commerce)

Strengths:

- Good compatibility across different devices
- Efficient data processing and fast cloud synchronization
- Convenient for multi-device or remote access setups

Weaknesses:

- Monthly subscription fees may not be affordable for small hardware stores
- Requires stable internet connectivity at all times
- Features may exceed what a micro-enterprise actually needs

Relevance:
This tool highlights the importance of creating a cost-efficient, locally hosted system—especially for businesses that cannot sustain ongoing monthly expenses or depend on internet-based services.

- **Technology Stack Rationale:**
  The project uses the following technologies:
  1. Frontend: Vue.js
  2. Backend: Laravel
  3. Database: MySQL
  4. Network Environment: Local Area Network (LAN)

Reasons for Choosing This Stack:

1. Vue.js
   - Lightweight and fast, suitable for simple inventory dashboards
   - Easy to learn and maintain, ideal for rapid prototyping
   - Component-based structure simplifies UI development

2. Laravel
   - Strong built-in security features (authentication, CSRF protection, hashing)
   - Clean API structure that fits the system's role-based access control
   - Excellent documentation and stable performance for local server setups
3. MySQL
   - Highly reliable relational database suited for structured records
   - Efficient for querying inventory and sales data
   - Easy to back up and restore for small business environments
4. Local Network Hosting
   - Ensures the system is only accessible within the store, improving security
   - No dependency on internet availability
   - Reduces operational costs compared to cloud-based solutions

Overall, the chosen tech stack supports the goals of creating a secure, efficient, and store-only system that is easy to maintain and scalable for future improvements.

2.2. Domain Analysis

Key Findings:
Research on small retail businesses consistently points out that manual processes—such as handwritten logs, physical receipts, and notebook-based tracking—lead to recurring issues:

- Stock inaccuracy due to the inconsistent updating of records
- Human errors in manual computations of sales and expenses
- Difficulty tracking fast-moving items, especially when transaction volume increases
- Time delays when searching for inventory information
- Limited visibility on sales trends and stock movement

A study by Reyes (2020) found that businesses switching to digital inventory tools experienced fewer discrepancies and improved decision-making due to real-time access to stock data. Meanwhile, Mendoza (2019) emphasized that digital systems reduce operational losses by minimizing errors and improving monitoring accuracy. Similar studies highlight how micro-businesses benefit from simple, accessible systems that integrate both sales and inventory tracking.

Gap Identification:
While many inventory applications already exist, most of them fall into two categories:

1. Too complex and feature-heavy for micro-enterprises
2. Too costly, requiring subscriptions or internet-based hosting
3. Not customizable for small hardware store workflows
4. Dependent on internet connectivity, which may not always be stable

This project fills those gaps by providing:

- A local-only, store-accessible system
- Straightforward workflows designed specifically for a hardware retail environment
- No subscription fees or external service requirements
- A beginner-friendly interface suitable for the store owner and staff

Thus, the proposed solution directly addresses the lack of affordable, simple, and secure tools for micro-businesses like XYM Mini Hardware.

3. Project Methodology

3.1. Development Process

The project will follow the **Agile Scrum Development Methodology**, which supports iterative development and continuous feedback from XYM Mini Hardware. Agile is suitable for this project because the owner's requirements may evolve as prototypes are tested, and the development team can adjust features through short, scheduled iterations.

Figure 3.1: Agile Scrum Development Cycle



Caption: The Agile Scrum Development Cycle emphasizes iterative progress through Sprints. For the XYM Mini Hardware system, we will use 2-week sprints to manage development.
Key ceremonies will include: Sprint Planning (determining the work for the next two weeks), Daily Scrums (15-minute sync-up meetings to discuss progress and roadblocks), Sprint Review (demonstrating the working increment to the owner for feedback), and Sprint Retrospective (for the team to reflect on the process and identify improvements). This allows for rapid feature delivery and ensures the final system aligns perfectly with the store's evolving needs.

3.2. Timeline & Milestones

Figure 3.2: Project Timeline & Key Milestones

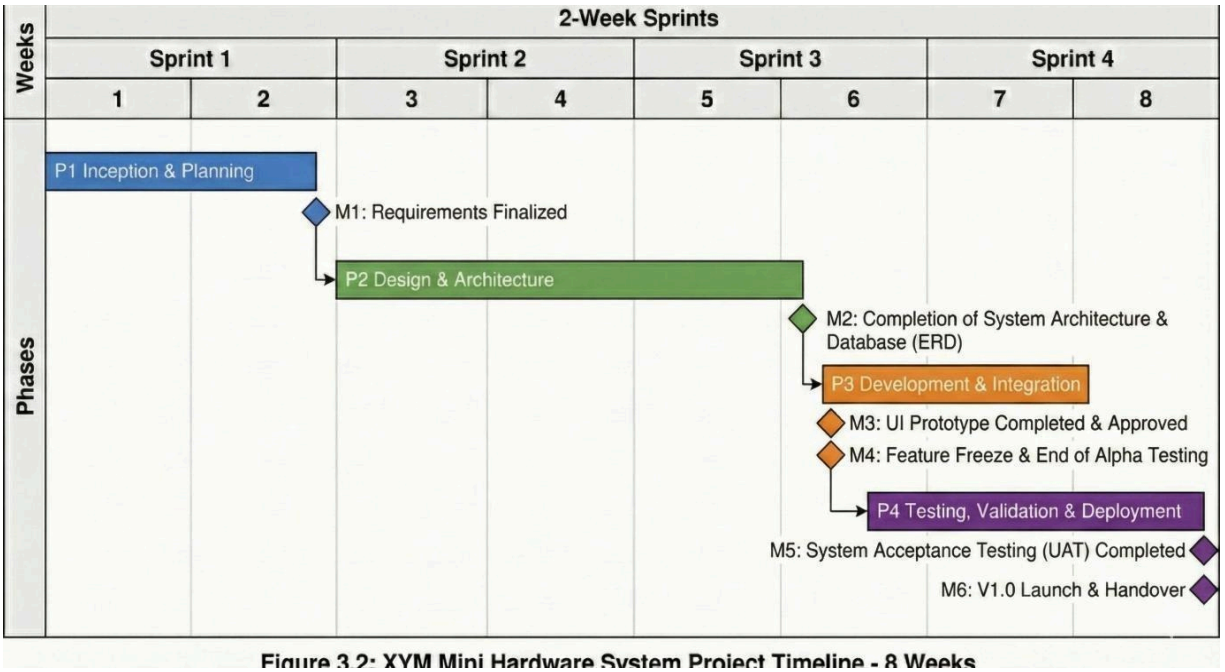


Figure 3.2: XYM Mini Hardware System Project Timeline - 8 Weeks

Caption: The project is structured into **four major phases** over a **2-month period**, utilizing 2- week Agile Sprints. Key milestones include:

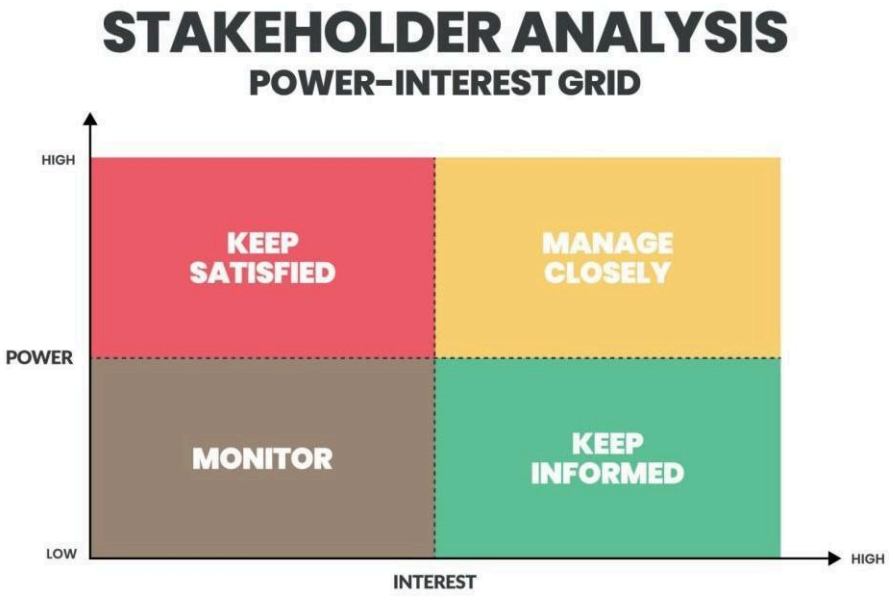
Phase ID	Phase Description	Key Milestones	Estimated Duration
P1	**Inception & Planning**	M1: Requirements Document Finalized & Approved	2 Weeks (Sprint 1)
P2	**Design & Architecture**	M2: Completion of System Architecture & Database (ERD)	4 Weeks (Sprints 2-3)
P3	**Development & Integration**	M3: UI Prototype Completed & Approved	3 Weeks (Sprints 3-4)

Phase ID	Phase Description	Key Milestones	Estimated Duration
		M4: Feature Freeze & End of Alpha Testing	
P4	**Testing, Validation & Deployment**	M5: System Acceptance Testing (UAT) Completed **M6: V1.0 Launch & Handover**	3 Weeks (Sprint 3-4)

3.3. Stakeholder Analysis

The success of the XYM Mini Hardware Inventory and Sales Monitoring System depends on managing the expectations and interests of several key parties. We have categorized them based on their influence and interest in the project outcome.

Figure 3.3.1: Stakeholder Power-Interest Matrix



Stakeholder Group	Description	Primary Interest	Influence (High/Medium/ Low)	Interest (High/Medium/ Low)	Management Strategy
Store Owner	The primary client and sponsor of the project; final	System accuracy, reduced errors, and access to	High	High	**Manage Closely:** Frequent communication , involving

	decision-maker.	sales reports.			them in the Sprint Reviews and UAT.
Inventory Auditor/Cashier	The main daily user who will input sales and inventory data.	System usability, speed, and reduction of manual computation workload.	High	High	**Manage Closely:** Prioritizing their workflow needs and seeking continuous feedback on the UI/UX.
Staff (View-Only User)	Staff member responsible for verifying physical stock against the system records.	Data reliability and ease of viewing inventory status.	Medium	Medium	**Keep Satisfied:** Ensure the reporting/viewing module meets their verification needs.
Development Team	The students are responsible for system development, testing, and documentation.	Successful delivery, meeting project goals, and achieving academic requirements.	High	High	**Manage Closely:** Clear requirements, adequate resources, and maintain project scope.
Mr. Klinth Jerald L. Matugas	Project instructor and academic advisor.	Adherence to software engineering standards, quality of documentation, and successful execution of the methodology.	Medium	High	**Keep Informed:** Providing regular status reports and academic deliverables.
Future Maintenance Team	(Hypothetical) Any future developer tasked with maintaining or updating the system.	Clean, well-documented, and modular source code.	Low	Low	**Monitor:** Ensure comprehensive documentation and adherence to coding standards.

Key Stakeholder Categories

- **Primary Stakeholders:**

  - Store Owner
  - Inventory Auditor/Cashier
  - Staff (View-Only User)
  - Development Team

- **Secondary Stakeholders:**

  - Mr. Klinth Jerald L. Matugas (Project Instructor)
  - Future Maintenance Team
  - Suppliers (Indirectly affected by more accurate ordering)

3.4. Initial Requirements Gathering

The initial requirements for the XYM Mini Hardware Inventory and Sales Monitoring System were gathered primarily through **semi-structured interviews** and **direct observation** of the store's current manual operations.

Methodology

1. **Semi-Structured Interviews:** A one-hour interview was conducted with the **Store Owner** and the **Cashier** to understand their daily workflow, existing pain points, expectations for an automated system, and non-negotiable needs (like retaining manual receipts for customers).
2. **Direct Observation:** The team spent half a day observing the process of logging sales, checking inventory levels, and preparing reports using the current handwritten logbooks. This was crucial for validating the identified pain points and understanding the exact data flow.
3. **Review of Existing Documents:** Sample handwritten sales logs, inventory withdrawal slips, and calculation sheets were analyzed to identify common error types and the structure needed for the digital database.

Key Insights that Shaped the Project

The initial gathering process confirmed the severity of the problem and directed the project toward a specific design focus:

- **The Primary Need is Accuracy, not Speed:** While the manual process is slow, the most significant risk cited by the owner was the **frequency of manual computation errors** and **inaccurate stock counts**, leading to lost revenue or wasted time during restocking.
- **Simplicity is Critical:** The owner and cashier are accustomed to a very simple, notebook-based system. Any complex interface or feature set would likely lead to resistance and non-adoption. This reinforced the decision for a **simple, local-only web application** focused purely on core inventory and sales tracking.
- **The 'Low-Stock' Alert is Essential:** The manual system's biggest failure point is detecting when an item needs restocking *before* it runs out. This insight directly led to the inclusion of a mandatory, real-time stock alert feature.

Figure 3.3: Key Insight from Initial Survey

1. Organization & Current System Overview

1. *Can you describe how daily operations are currently managed (sales, inventory, deliveries)?*

   Twice a year we conduct an inventory. Sales are recorded in delivery receipts. Customer orders have corresponding delivery receipts, which are given to our delivery personnel. Each order also has a withdrawal slip for the warehouse indicating what will be loaded if the items are not in the store. Before giving a withdrawal slip, the owner signs it. The checker won't release anything without the owner's or assistant's signature. That is how inventory is monitored. For sales, delivery receipts cover deliveries, and sales invoices cover walk-in customers.

2. *What systems or tools are you using now (POS, spreadsheets, manual logs, etc.)?*

   We use manual logs. For example, petty cash and sales are recorded on paper. If there is a cash shortage, we compare cash sales with sales indicated in delivery receipts and invoices. We total it daily.

3. *Who mainly uses these systems (cashiers, owner, staff)?*

   The assistant.

4. *What do you find effective about your current system?*

   The owner has always used manual systems. Actual inventory based on his experience is the basis for customer orders. Computers sometimes show discrepancies compared to actual stocks, which could mean damages, over-releasing stock, or even theft. Manual monitoring helps identify these issues. For the owner, using manual is more convenient and faster for customers, as computerized processes can cause delays. However, the owner also recognizes that a digital setup could provide faster stock monitoring and better inventory tracking in the future.

Caption: *Data collected from the store owner and cashier indicated that* ***"Frequent Errors in Sales Computation/Stock Counts"*** *was the single most impactful pain point, affecting daily operations* ***(80% impact score)****. This key insight confirmed that the core value proposition of the system must be automation to ensure data integrity and accuracy.*

4. System Specification

4.1. User Personas & Stories

Based on the initial interviews and observation, three distinct user roles were identified for the XYM Mini Hardware Inventory and Sales Monitoring System.

User Personas

Persona 1: The Owner/Decision Maker

Detail	Description
Name	Mr. Cua
Role	Store Owner (System Administrator)
Goal	Monitor overall business performance, ensure inventory accuracy, and make informed decisions based on reliable sales and stock data.
Pain Points	High risk of financial loss due to calculation errors; slow process of reviewing comprehensive reports (currently manual); security concerns regarding external access.
System Expectation	A secure, *local-only* system that provides quick, trustworthy sales and stock summaries without disrupting the current manual receipt process.
Influence	**High**—Approves all final system changes and is the primary reviewer of reports.

Persona 2: The Inventory Auditor/Cashier

Detail	Description
Name	Mrs. Joy and Mrs. Jonalyn
Role	Cashier (Handles sales and inventory data entry) and Inventory Auditor (Handles stock movement and verification)
Goal	Reduce daily workload, eliminate errors in sales totaling, and quickly retrieve stock information for customer inquiries.
Pain Points	Overload from manual computation and report preparation for the owner (especially BIR taxes); difficulty quickly identifying low stock; bearing the burden of discrepancy investigation.
System Expectation	A fast, intuitive interface for recording sales and updates, automating all computations, and providing immediate low-stock alerts.
Influence	**High**—Is the main source of data input and directly affected by the system's usability.

Persona 3: The Staff

Detail	Description
Name	Mr. Ramram
Role	Staff (Handles daily stock handling and monitoring)
Goal	Monitor inventory status and stock movements to support accurate daily operations and reporting.
Pain Points	Reliance on handwritten withdrawal slips for verification; limited real-time visibility of stock levels and reorder alerts.
System Expectation	View-only access to the dashboard and inventory lists to check current stock levels, low-stock alerts, and stock movement history.
Influence	**Medium** — Supports data accuracy and monitoring but does not modify or control system data.

User Stories

The following user stories define the core functionality required for the system, organized by user role:

Owner (System Administrator)

1. As the **Owner**, I want to **log into the system using a unique password**, so that I can ensure only authorized personnel can access business data. (FR-1)
2. As the **Owner**, I want to **view a real-time summary dashboard of daily sales and current inventory totals**, so that I can quickly assess the business's performance without manual calculations.
3. As the **Owner**, I want to **review detailed historical sales reports by date range**, so that I can monitor sales growth and trends.
4. As the **Owner**, I want the system to be **accessible only on the local store network**, so that the business data is secure and protected (Goal 3.1).

Inventory Auditor/Cashier

5. As the **Inventory Auditor/Cashier**, I want to **input sales transactions quickly by item name and quantity**, so that I can minimize manual recording and focus more on customer service.
6. As the **Inventory Auditor/Cashier**, I want the system to **automatically calculate the total sales value per transaction**, so that I can reduce human errors and speed up the checkout process.
7. As the **Inventory Auditor/Cashier**, I want to **update stock levels when new inventory arrives**, so that the system always reflects accurate item counts.
8. As the **Inventory Auditor/Cashier**, I want to **receive real-time visual alerts when stock levels fall below the minimum threshold**, so that I can flag items for reordering and prevent stock outs.

Staff

9. As the **Staff**, I want **view-only access to current stock quantities**, so that I can verify physical inventory against system records without modifying data.
10. As the **Staff**, I want to **view the stock-in and stock-out history**, so that I can cross- check inventory movements with signed withdrawal slips.

4.2. Functional Requirements

ID	Feature / Module	Description	Linked Goal/Stories
FR- 1	**User Management & Access Control**	The system shall enforce Role-Based Access Control (RBAC) with three privileges (Owner, Cashier & Inventory Auditor, Staff) and manage user credentials for secure login.	Goal 3.2, US 1, US 4
FR- 2	**Sales Transaction Processing**	The system shall allow the Cashier & Inventory Auditor to record sales by selecting items and quantity, automatically calculating the total sale price, and decrementing stock counts accordingly.	Goal 2.1, US 6, US 7

FR- 3	**Payment Recording & Tracking**	The system shall allow the Cashier to specify the payment type (Cash, GCash, Cheque) for each sale transaction.	US 10
FR- 4	**Inventory Item Management**	The system shall allow the Owner to add, edit, or delete inventory items, including setting cost price, selling price, and a customizable minimum stock threshold.	Goal 1.1, US 8
FR- 5	**Stock In/Out Recording**	The system shall allow the Cashier & Inventory Auditor to log manual stock adjustments (additions from delivery or subtractions due to damages/returns) and update the item quantity.	Goal 1.1, US 8
FR- 6	**Low Stock Alerting**	The system shall automatically detect if an item's quantity falls below its minimum threshold and display a persistent, visible alert on the relevant user dashboards (Owner).	Goal 1.2, US 9
FR- 7	**Real-Time Summary Dashboard**	The system shall display a dashboard view showing real-time, aggregated data, including Total Sales for the day and a list of low-stock items.	US 2, US 6
FR- 8	**Historical Reporting & Export**	The system shall allow the Owner to generate and filter comprehensive sales and stock movement reports by date range and category, and export these reports (e.g., as CSV) for external use (e.g., BIR tax preparation).	US 3, US 12
FR- 9	**View-Only Inventory Module**	The system shall provide a dedicated, non-editable module accessible to the staff role, displaying current stock quantities and a log of recent inventory movements.	US 11, US 12

4.3. Non-Functional Requirements (NFRs)

Category	Requirement	Linked Goal/Objective
Performance	**NFR-P1:** The UI shall respond to user inputs (e.g., form submissions, navigation) in under 2 seconds.	Objective 1.2
Performance	**NFR-P2:** Inventory queries for displaying the main stock table and low stock alerts shall complete within 3 seconds.	Objective 1.2
Usability	**NFR-U1:** The system shall be intuitive enough for a non-technical user (Cashier & Inventory Auditor) to operate with less than 15 minutes of training.	Objective 2.2
Usability	**NFR-U2:** The data entry process for a typical sales transaction should require no more than three clicks/taps after logging in.	Objective 2.2
Reliability	**NFR-R1:** The system shall have 99.9% uptime during business hours (assuming stable local server hardware).	-
Security	**NFR-S1:** The system shall reject any access request originating from outside the local area network (LAN).	Goal 3.1
Security	**NFR-S2:** All user credentials and sensitive business data shall be stored using industry-standard hashing and encryption techniques (e.g., Laravel's built-in security features).	Goal 3
Maintainability	**NFR-M1:** The codebase shall be modular and well-documented (using Vue.js components and Laravel Eloquent models) to allow future maintenance teams to implement V2.0 features efficiently.	-

4.4. System Architecture & Design

- 4.4.1. Use Case Diagram:

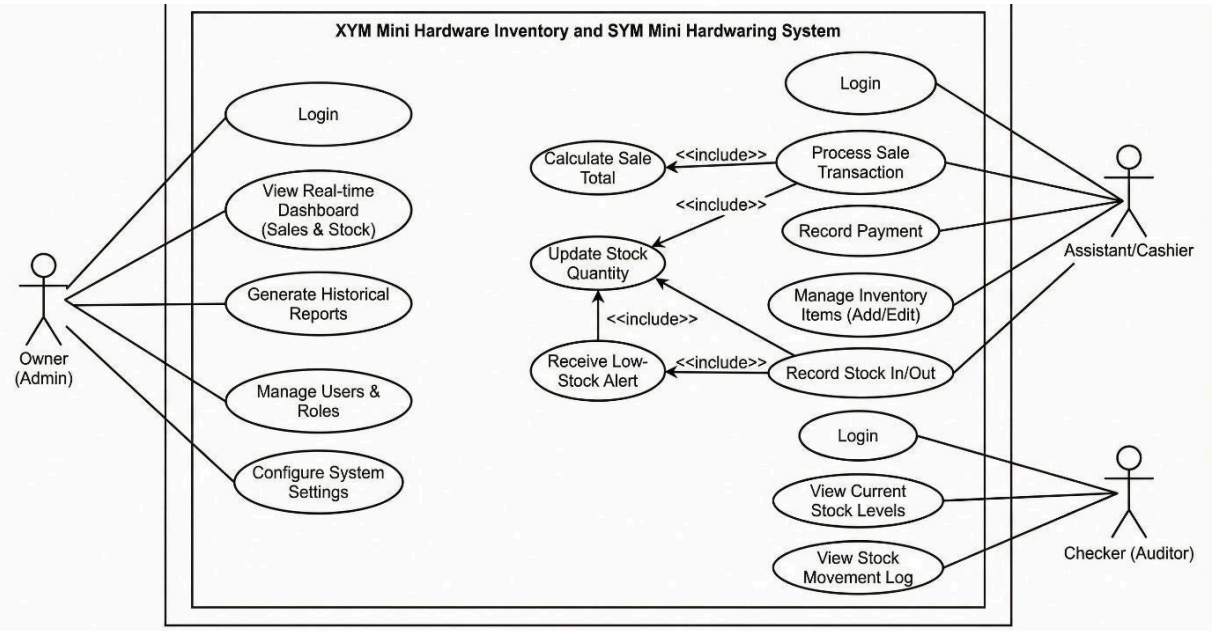


Figure 4.4.1: Use Case Diagram for XYM Mini Hardware System

Figure 4.4.1. Use Case Diagram. This diagram illustrates the functional boundaries of the system and the roles of the three primary actors: **Owner** (Manages Users & Views Reports), **Cashier and Inventory Auditor** (Manages Sales & Inventory Updates), and **Staff** (Views Stock Status). The core system use cases revolve around authentication, managing inventory, recording sales, and generating reports.

- 4.4.2. System Architecture Diagram:

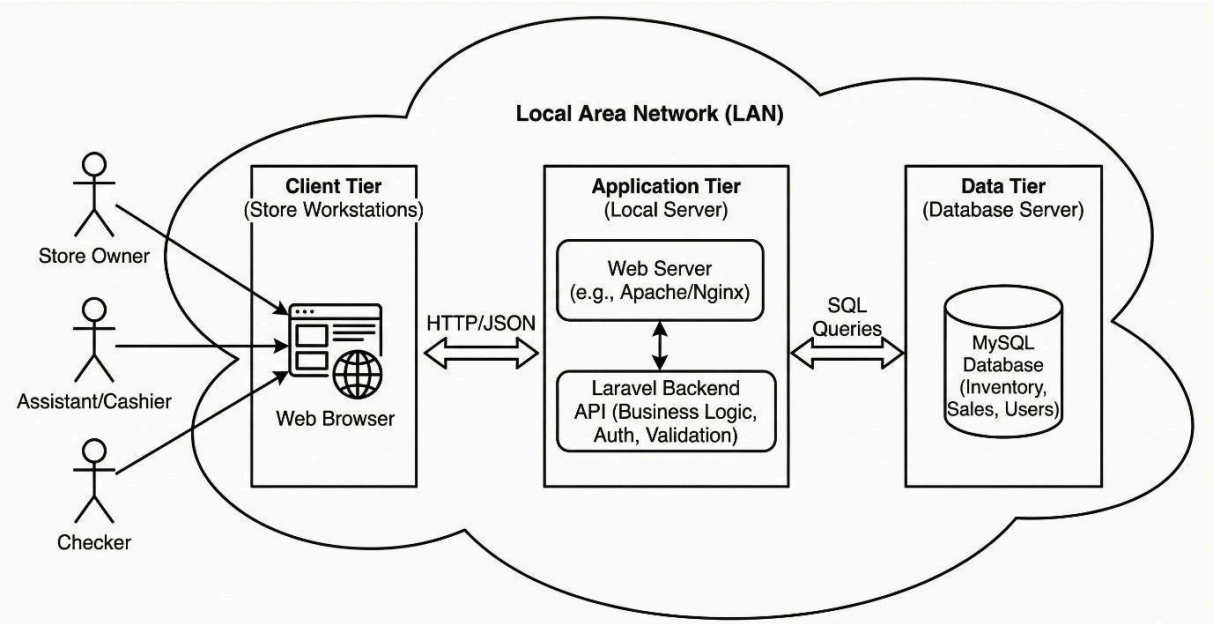


Figure 4.4.2: System Architecture Diagram for XYM Mini Hardware System

Figure 4.4.2. System Architecture Diagram. The proposed architecture follows a classic three-tier model, hosted entirely within the store's Local Area Network (LAN): 1) **Presentation Tier (Frontend):** Vue.js provides the responsive user interface. 2) **Application Tier (Backend):** Laravel acts as the API server, handling business logic, authentication, and data validation. 3) **Data Tier (Database):** MySQL is the reliable relational database for storing all inventory, sales, and user records. All communication occurs locally.

- 4.4.3. Class Diagram:

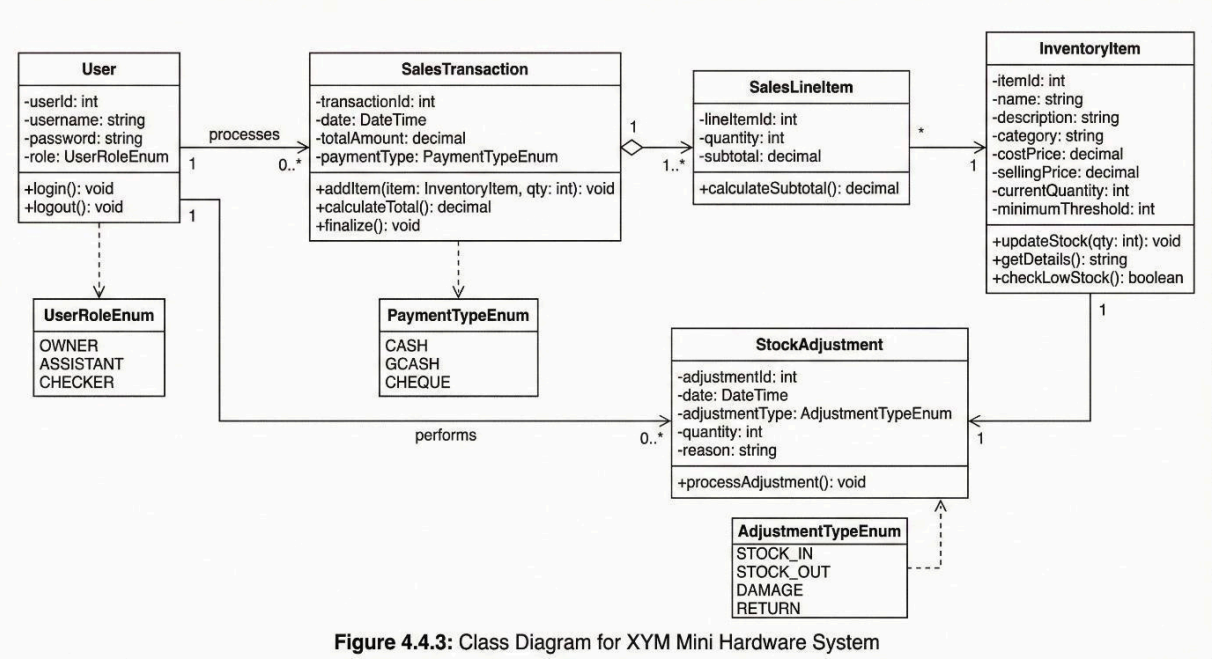


Figure 4.4.3: Class Diagram for XYM Mini Hardware System

Figure 4.4.3. Class Diagram. This UML diagram shows the main entities and their relationships within the system's business logic. Key classes include User (with roles), InventoryItem (tracks stock levels and thresholds), SalesTransaction (records sales details), and StockAdjustment (logs stock changes not due to sales). This structure ensures data integrity and modularity.

- 4.4.4. Entity-Relationship Diagram (ERD

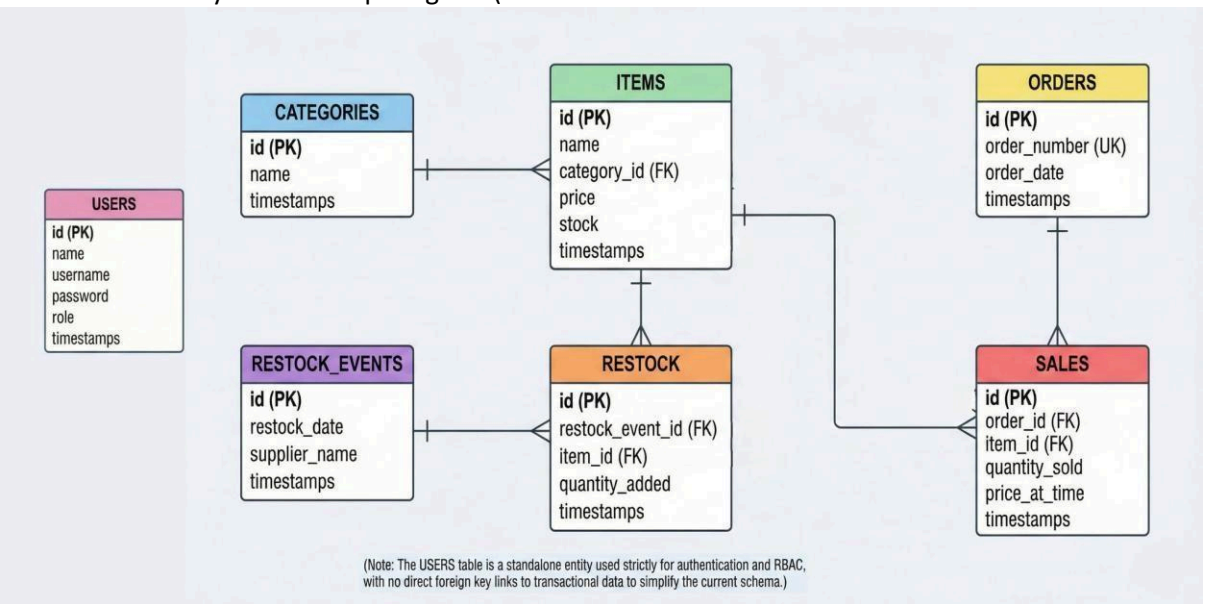


Figure 4.4.4. Entity-Relationship Diagram (ERD). The diagram illustrates the structure of the MySQL database used for the XYM Mini Hardware Inventory and Sales Monitoring System. The **CATEGORIES** table maintains a one-to-many relationship with **ITEMS,** organizing products by category. The **ITEMS** table serves as the central inventory entity, connected to **SALES** for recording sold items and to **RESTOCK** for tracking added stock quantities. Sales transactions are grouped under **ORDERS**, which holds a one-to-many relationship with the SALES table. Inventory replenishment activities are documented through **RESTOCK_EVENTS**, which link to individual item restock records. The USERS table functions as a standalone entity for authentication and role-based access control. Overall, the ERD supports accurate tracking of inventory levels, stock movements, and sales transactions.

5. Feasibility & Risk Analysis

5.1. Technical Feasibility

- **Technology Maturity:** The chosen technology stack—**Vue.js (Frontend), Laravel (Backend), and MySQL (Database)**—are all mature, widely adopted, and well-supported platforms. Vue.js offers a low learning curve and high performance suitable for a straightforward application, while Laravel provides robust built-in security and MVC structure necessary for API development. MySQL is an industry-standard, reliable relational database, ideal for structured inventory and sales records. Critically, the choice of **Local Area Network (LAN) hosting** leverages mature, stable network technology and bypasses reliability issues associated with consumer internet in the store environment (Goal 3.1).

- **Team Skills:** The development team possesses proficiency in HTML, CSS, JavaScript, and database management, covering the foundational elements of the project. While the team may require a short ramp-up period for the specific framework nuances of Vue.js and Laravel, these are addressed as a mitigation strategy (R-3) within the project's aggressive 8- week timeline. The project's focused scope (no external APIs, no advanced analytics) limits the complexity, making it achievable with the team's current and augmented skill set.

- **Proof of Concept:** Technical viability is proven by the simplicity of the core functions: 1) **Database Transaction Integrity:** The relational structure (ERD) ensures accurate stock decrement upon sale, which is a standard, repeatable database transaction. 2) **Local Deployment:** Hosting the application on a local server is technically straightforward and aligns perfectly with the security non-functional requirements (NFR-S1). 3) **Role-Based Access:** Laravel provides out-of-the-box features for implementing RBAC, verifying that the multi-level access control (Owner, Cashier & Inventory Auditor, Staff) is achievable.

5.2. Risk Register

Risk ID	Description	Probability	Impact	Mitigation Strategy
R-1	Local server hardware failure causes data loss or system downtime.	Medium	High	Implement automated daily database backups to external storage (e.g., a flash drive or another local PC). Have a clear procedure for quick system re-installation.

R-2	Resistance from staff/owner to adopt the digital system due to complexity.	High	Medium	Prioritize simplicity and usability (NFR: Usability); conduct thorough, simple training sessions; and ensure the system solves a critical pain point (computation errors).
R-3	Team lacks experience with Laravel/Vue.js leading to development delays.	Medium	Medium	Allocate extra time in Sprint 1 for training; focus on core, well-documented features; and use simplified frameworks/libraries.
R-4	Inaccurate data input by the Cashier & Inventory Auditor compromises inventory records.	High	Medium	Implement data validation checks on all inputs (e.g., quantity must be positive, prices must be numeric); use drop-down item selection to minimize typos; implement audit logs for stock changes.
R-5	The aggressive 8-week timeline causes feature creep or incomplete testing.	Medium	High	Strictly adhere to the scope defined in Section 1.5; rigorously track progress in daily scrums; prioritize completing core functional requirements (FR-1 through FR-7) over all others.

6. Testing, Validation & Deployment

6.1. Test Strategy

To ensure the quality, reliability, and usability of the XYM Inventory and Sales Monitoring System, a multi-level testing strategy was applied. This strategy includes **Unit Testing**, **Integration Testing**, and **User Acceptance Testing (UAT)** to validate both technical correctness and user satisfaction.

Unit Testing

Unit testing was performed on isolated frontend components and backend business logic to ensure individual modules function as intended. Frontend components developed using Vue.js were tested to verify correct data rendering and user interaction behavior, while backend logic implemented in Laravel was tested to ensure accuracy and reliability of core system functions. Key focus areas included:

- Accuracy of automated sales computation (**FR-2**)
- Correct inventory increment and decrement logic for sales and restocking (**FR-4**, **FR-5**)
- Enforcement of Role-Based Access Control to prevent unauthorized access (**FR-1**)

Integration Testing

Integration testing focused on validating communication between the system's layers, ensuring seamless interaction between the frontend, backend, and database. API endpoints were tested to confirm that data submitted from the Vue.js interface is correctly processed and stored by the Laravel backend. This phase included:

- Verifying end-to-end sales transaction workflows, ensuring that recorded sales correctly trigger inventory deductions
- Testing real-time low-stock alert functionality from database updates to dashboard display (**FR-6**, **Objective 1.2**)
- Ensuring consistent data synchronization across user dashboards

User Acceptance Testing (UAT)

User Acceptance Testing (UAT) was conducted to evaluate whether the system meets the operational needs, expectations, and satisfaction of its intended users in a real store environment. The UAT involved key stakeholders of XYM Mini Hardware, including the **Store Owner**, **Cashier/Inventory Auditor**, and **Store Staff**, representing actual system users.

UAT Activities Included:

- Allowing users to perform core operational tasks such as processing sales transactions, monitoring inventory levels, viewing low-stock alerts, and generating daily sales reports
- Observing ease of navigation, clarity of the user interface, and role-based access control
- Collecting structured feedback through a post-implementation survey using a **5-point Likert scale** to measure usability, accuracy, and data security

Success Criteria Included:

- Users completed assigned tasks without external assistance
- No critical system errors occurred during actual usage
- The system achieved an average usability and satisfaction rating of **at least 4.0 out of 5**

Spreadsheet link: https://tinyurl.com/xymspreedsheet

UAT Results Summary

The UAT survey was conducted on January 02, 2026, involving three (3) respondents with different system roles:

- The **Store Owner (System Administrator & Decision Maker)** reported an individual average rating of **4.8**, indicating strong agreement that the system improved sales processing speed, accuracy, inventory monitoring, reporting, and data security.

- The **Cashier/Inventory Auditor (Primary Daily User)** provided an individual average rating of **4.7**, reflecting high satisfaction with transaction efficiency, system usability, inventory alerts, and report reliability.

- The **Store Staff (View-Only User)** reported an individual average rating of **4.5**, confirming ease of use and confidence in data security despite limited system access.

Overall, all respondents rated the system within the **"Strongly Agree"** interpretation range, exceeding the predefined success criteria and confirming that the system is acceptable for operational deployment.

Summary Link: https://tinyurl.com/XYMlikertperindividual

6.2. User Validation Plan

The User Validation Plan was designed to formally assess whether the **XYM Inventory and Sales Monitoring System** meets user expectations in terms of **usability, accuracy, and ease of operation** after deployment.

Method:

User validation was performed through a **structured usability testing session**, followed by a **post-test survey administered via Google Forms**. The survey used a **5-point Likert Scale** to measure user satisfaction, ease of use, system accuracy, and perceived data security.

During the usability testing session, participants were asked to perform system tasks based on their assigned roles, including:

- Recording a standard sales transaction
- Viewing and generating daily sales reports
- Monitoring inventory levels and responding to low-stock alerts

Responses collected through Google Forms were automatically recorded and summarized for analysis.

Success Criteria:
The system was considered successfully validated if it met the following criteria:

1. All participants completed their assigned usability tasks **without external assistance**
2. The system achieved a **mean satisfaction score of at least 4.0 out of 5** across all usability-related questions, validating **NFR-U1 (Ease of Learning)** and **NFR-U2 (Minimal Interaction Steps)**

- **Likert Scale Interpretation:**

Score	Interpretation	Meaning
5	Strongly Agree	User highly satisfied; system is intuitive and easy to use.
4	Agree	User satisfied; minor usability issues may exist.
3	Neutral	System is acceptable but neither easy nor difficult to use.
2	Disagree	User dissatisfied; system requires effort or causes frustration.
1	Strongly Disagree	User highly dissatisfied; system is unusable or complex.

6.3. Deployment Plan

- **Environment:** The XYM Inventory and Sales Monitoring System is deployed on a dedicated local server PC within the existing Local Area Network (LAN) of XYM Mini Hardware. The system operates offline and is accessible only within the store premises, ensuring compliance with NFR-S1 (System Security) and Goal 3.1 by preventing unauthorized external access.

- **Deployment Process:** The deployment followed a **manual on-site installation process** to ensure full control over system configuration and security. The process included the following steps:

1. **Server Setup**
   Installation and configuration of the required server environment, including **PHP**, **MySQL**, and **Apache/Nginx**, on the local server PC.
2. **Application Deployment**
   Transfer and setup of the finalized and fully tested **Laravel (backend)** and **Vue.js (frontend)** application codebase on the local server.
3. **Database Configuration**
   Creation of the MySQL database, execution of initial migrations, and import of existing inventory and product records.
4. **Network Configuration**
   Assignment of a **static local IP address** to the server to ensure consistent access, along with firewall configuration to **restrict external and internet-based traffic**.
5. **Handover and Training**
   A final system handover and training session was conducted for the **store owner**, covering system operation, basic troubleshooting, and **daily and weekly database backup procedures**, serving as a mitigation strategy for **Risk R-1 (Data Loss)**.

7. Conclusion & Future Work

The **XYM Mini Hardware Inventory and Sales Monitoring System (Version 1.0)** successfully addresses the primary challenges of **manual sales computation errors** and **inventory inconsistencies**, which were previously caused by the use of handwritten logbooks. The system fulfills its core objective of improving transaction accuracy while reducing staff workload and operational delays.

By utilizing a **secure, local-only system architecture** built with **Vue.js, Laravel, and MySQL**, the project delivers a **cost-effective, reliable, and data-secure solution** tailored specifically to the operational needs of a micro-business environment such as XYM Mini Hardware. The system demonstrated strong performance by achieving high sales calculation accuracy and providing **real-time inventory updates**, ensuring more reliable stock monitoring and reporting.

The intentionally **limited system scope** (as defined in Section 1.5) was critical to meeting the project's **8-week development timeline**. Despite this constraint, the successful deployment and positive User Acceptance Testing results indicate that the system is **operationally viable and well-received** by its intended users. These results establish a solid foundation for future system expansion.

Future Work (V2.0 Features)

To further enhance system functionality and long-term business value, the following features are recommended for future development:

1. **Supplier and Purchase Order Tracking**
   Implement a supplier management module that records supplier details, tracks purchase orders (POs), and automatically updates inventory levels upon PO receipt confirmation, addressing the owner's initial interest in supplier-based stock monitoring.

2. **Mobile View for Owner Reports**
   Develop a secure, view-only mobile application or mobile-optimized web interface that allows the store owner to remotely monitor sales summaries and inventory status, provided that external access security controls aligned with **Goal 3.1** can be robustly implemented.

3. **Basic Sales Analytics**
   Introduce historical sales analytics such as monthly sales comparisons, trend visualization, and identification of fast-moving and slow-moving items to support more informed business decision-making.

4. **Barcode and QR Code Integration**
   Integrate barcode or QR code scanning functionality to speed up sales transactions and physical inventory counting, further improving operational efficiency and data accuracy.

5. **Digital Receipt Generation**
   Provide an optional digital receipt feature, such as a printable PDF transaction summary, to streamline the checkout process while still accommodating the store's current preference for manual receipts.

8. References

Academic and Industry Sources:

- **Beck, K., & Andres, C.** (2004). *Extreme programming explained: Embrace change* (2nd ed.). Addison-Wesley Professional. (Reference for Agile/Scrum Methodology)
- **Mendoza, L. C.** (2019). The impact of digital inventory management systems on micro, small, and medium enterprises (MSMEs) in the Philippines. *Philippine Journal of Business and Economics*, *58*(1), 45-62. (Supports Domain Analysis and problem with manual processes)
- **Reyes, A. B.** (2020). *Inventory accuracy and financial performance in local hardware retail stores*. Unpublished master's thesis, University of Mindanao, Davao City, Philippines. (Supports Domain Analysis and local context)
- **Schwaber, K., & Sutherland, J.** (2020). *The Scrum guide*. Scrum.org. (Primary source for Scrum framework)

Technical Documentation:

- **Laravel Framework Documentation.** (n.d.). Retrieved from https://laravel.com/docs (Documentation for Backend/API framework)
- **MySQL Documentation.** (n.d.). Retrieved from https://dev.mysql.com/doc/ (Documentation for Database Tier)
- **Vue.js Documentation.** (n.d.). Retrieved from https://vuejs.org/guide/ (Documentation for Frontend framework)

9. Appendices

- **Appendix A:** Full Interview Questions and Answer
  Link: https://tinyurl.com/xymquestions

Interview Questions for XYM Mini Hardware

1. Organization & Current System Overview

1. *Can you describe how daily operations are currently managed (sales, inventory, deliveries)?*

   Twice a year we conduct an inventory. Sales are recorded in delivery receipts. Customer orders have corresponding delivery receipts, which are given to our delivery personnel. Each order also has a withdrawal slip for the warehouse indicating what will be loaded if the items are not in the store. Before giving a withdrawal slip, the owner signs it. The checker won't release anything without the owner's or assistant's signature. That is how inventory is monitored. For sales, delivery receipts cover deliveries, and sales invoices cover walk-in customers.

2. *What systems or tools are you using now (POS, spreadsheets, manual logs, etc.)?*

   We use manual logs. For example, petty cash and sales are recorded on paper. If there is a cash shortage, we compare cash sales with sales indicated in delivery receipts and invoices. We total it daily.

3. *Who mainly uses these systems (cashiers, owner, staff)?*

   The assistant.

4. *What do you find effective about your current system?*

   The owner has always used manual systems. Actual inventory based on his experience is the basis for customer orders. Computers sometimes show discrepancies compared to actual stocks, which could mean damages, over-releasing stock, or even theft. Manual monitoring helps identify these issues. For the owner, using manual is more convenient and faster for customers, as computerized processes can cause delays. However, the owner also recognizes that a digital setup could provide faster stock monitoring and better inventory tracking in the future.

2. Sales & Transactions

5. *How do you record sales transactions (POS, receipts, manual logbook)?*

   Manual logbook.

6. *Do you encounter issues with tracking sales or generating reports?*

   Yes. Sometimes there are discrepancies that cannot be avoided. We cannot always confirm daily losses; we only rely on the checker to record stock in and out of the warehouse. Sales growth also varies—sometimes high, sometimes low.

7. *What payment methods do customers usually use (cash, GCash, PayMaya, credit)?*

   GCash and cheque.

3. Inventory Management

8. *How do you track available stocks and monitor shortages?*

   Checkers record incoming and outgoing stock. So far, there has been no major loss, but sometimes excess stock is received. Withdrawal slips are marked to ensure exact amounts are released.

9. *Do you face difficulties in knowing when to reorder supplies?*

   Yes, sometimes. Since we rely on manual monitoring, there are instances when we only realize that stock is low once customers order. If suppliers also face delays, we experience stockouts.

10. *Have there been cases of overstocking or stockouts? If yes, how often?*

    Yes. We once ordered products that customers didn't buy, and it lasted for over a year until we slowly disposed of it. For stockouts, delays happen when suppliers run out of stock, and even at the factories, trucks line up for supplies.

4. Information Flow & Reporting

11. *How do you generate reports (sales summary, inventory reports, expenses)?*

    Manual.

12. *Who reviews these reports, and how often?*

    The assistant prepares them, but the owner reviews them. It's important to monitor inventory levels per supplier to avoid running out of stock.

13. *Is there a need for faster or more automated reporting?*

    Yes. Faster reporting would save time and make it easier to monitor sales and inventory trends. Automated reporting could also reduce human error in calculations.

5. Customer & Supplier Management

14. *Do you maintain records of frequent customers or suppliers?*

    Yes, but only for customers with credits.

15. *How do you usually communicate with suppliers (phone, email, messaging apps)?*

    Through phone calls.

16. *Would a system that tracks supplier orders and deliveries be helpful?*

    Yes, it would be helpful, but the owner still prefers to keep inventory tracking manual.

6. Challenges & Pain Points

17. *What problems do you encounter in your current system?*

- (e.g., slow transaction time, errors in records, lost receipts, manual work overload)

We usually encounter errors in records due to human error, especially when totaling sales. Manual work also creates an overload, particularly when preparing reports for BIR tax purposes. However, we do not lose receipts because we use triplicate copies, and transactions are not delayed since we have three cashiers to handle customers.

18. *Are there times when operations are delayed because of system or manual process issues?*

Not due to the manual process itself, but delays happen when staffs or workers are absent. Orders pile up, and if they go beyond 11:00 AM, they are carried over to the next day.

7. Desired Features & Improvements

19. *If given a new system (app or website), what features would you like it to have?*

- (e.g., inventory alerts, sales reporting, digital receipts, supplier management)

If given a new system, the owners would like it to have features for inventory monitoring, supplier tracking, and stock monitoring.

20. *Do you prefer a mobile app, a website, or both for managing business operations?*

The owner prefers a mobile application.

21. *Should the system be accessible only inside the store, or also remotely (owner access anywhere)?*

Only inside the store.

8. Future Goals

22. *Do you plan to expand your store or add branches in the future?*

No. The store is already enough to meet the owner's needs for food, clothing, and financial support for his family.

23. *If yes, would you need a system that can handle multiple branches?*

The owner says no.

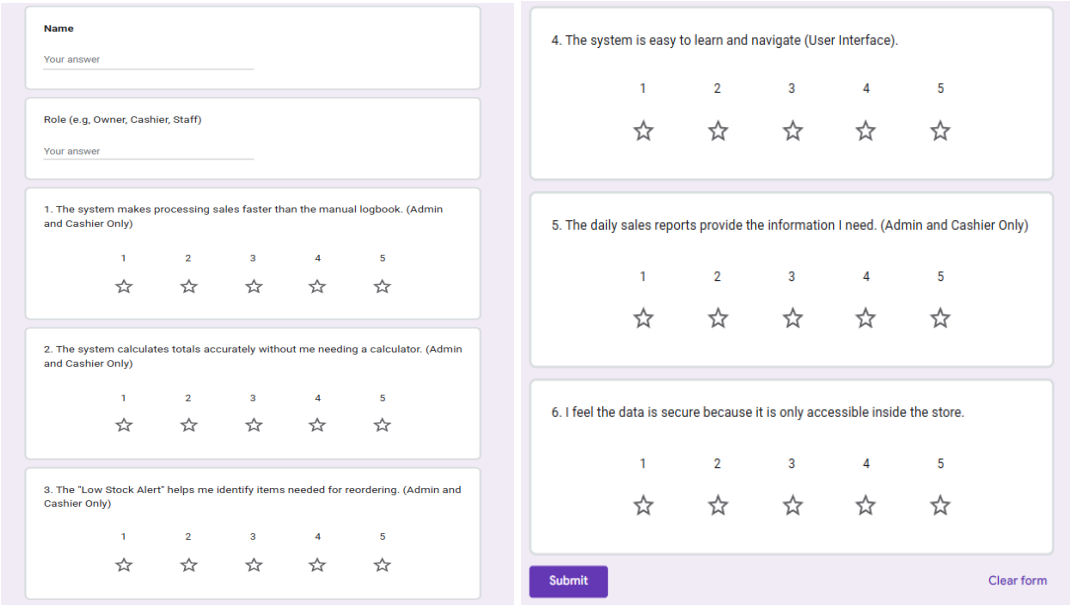
24. *How willing are you to adopt more digital solutions to improve efficiency?*

It would be better, but for now, we prefer manual solutions. Maybe in the future, the next generation of owners will apply digital solutions.

- **Appendix B:** Full Set of User Stories Link: https://tinyurl.com/xymuserstories

User Story		
User Persona	I want to <Goal/Objective>	So that <benefit/result/some reason>
Name: Mr. William Cua **Address:** P-3 Baan Butuan City **Position:** Owner of XYM Mini Hardware **Problems Encountered:** Errors in records due to manual inventory tracking Difficulty in monitoring stock levels, shortages, and overstocking Overstocking of items that don't sell fast, tying up capital for a long time. Overstocking when a buyer places a large order, but later cancels after the store already ordered from the supplier. Manual workload overload for the assistant, especially when preparing reports. No automated alerts for low stock, causing late realization of shortages. Customer credit records are harder to manage and may be misplaced since they are only logged manually. Limited visibility on sales and inventory trends (fast-moving vs. slow-moving items). Risk of human error in totaling sales, which makes financial records less reliable.	I want to generate simple but accurate inventory reports that are easy to review and compare with manual records.	So that I can minimize human errors, save time on BIR and tax reporting, and still retain control over manual sales transactions for faster customer service.
	I want to have a system that focuses on monitoring inventory levels, including alerts for low stock and overstocking.	So that I can prevent product shortages, avoid waste from unsold items, and always keep the right balance of stocks in the warehouse.
	I want the system to automatically calculate and display daily sales totals based on recorded inventory transactions.	So that I can know exactly how much sales the store makes each day, reduce manual computation, and maintain accurate financial tracking without slowing down customer service.
	I want the system to store customer credit records digitally	So that I can easily track which customers still owe payments, reducing the chance of forgetting or losing records.
	I want the system to allow my assistant limited access	So that I can delegate tasks like adding or checking stock without risking unauthorized changes.
	I want the system to provide inventory trend insights (fast-moving vs. slow-moving items)	So that I can prioritize which products to stock more and avoid wasting money on slow-moving items.
	I want an app that can only be accessed inside our store.	So that I can ensure business data stays secure and operations remain controlled only within the hardware store.

- **Appendix C:** Full Survey Questions and Raw Data
- Link:https://docs.google.com/forms/d/e/1FAIpQLSd1VxdQOV3ZP3oiRs8bwF5PDzAbQvPttFJ8mXefDNU4x2jwoQ/viewform



Name

William Cua

Role (e.g, Owner, Cashier, Staff)

Owner

1. The system makes processing sales faster than the manual logbook. (Admin and Cashier Only)

1 2 3 4 5

2. The system calculates totals accurately without me needing a calculator. (Admin and Cashier Only)

1 2 3 4 5



3. The "Low Stock Alert" helps me identify items needed for reordering. (Admin and Cashier Only)

1 2 3 4 5

4. The system is easy to learn and navigate (User Interface).

1 2 3 4 5



5. The daily sales reports provide the information I need. (Admin and Cashier Only)

1 2 3 4 5



6. I feel the data is secure because it is only accessible inside the store.

1 2 3 4 5



Name

JOY

Role (e.g, Owner, Cashier, Staff)

Cashier

1. The system makes processing sales faster than the manual logbook. (Admin and Cashier Only)

1 2 3 4 5



2. The system calculates totals accurately without me needing a calculator. (Admin and Cashier Only)

1 2 3 4 5



3. The "Low Stock Alert" helps me identify items needed for reordering. (Admin and Cashier Only)

1 2 3 4 5



4. The system is easy to learn and navigate (User Interface).

1 2 3 4 5



5. The daily sales reports provide the information I need. (Admin and Cashier Only)

1 2 3 4 5



6. I feel the data is secure because it is only accessible inside the store.

1 2 3 4 5

Name

RemRem

Role (e.g, Owner, Cashier, Staff)

Staff

1. The system makes processing sales faster than the manual logbook. (Admin and Cashier Only)

1 2 3 4 5

2. The system calculates totals accurately without me needing a calculator. (Admin and Cashier Only)

1 2 3 4 5



3. The "Low Stock Alert" helps me identify items needed for reordering. (Admin and Cashier Only)

1 2 3 4 5



4. The system is easy to learn and navigate (User Interface).

1 2 3 4 5



5. The daily sales reports provide the information I need. (Admin and Cashier Only)

1 2 3 4 5

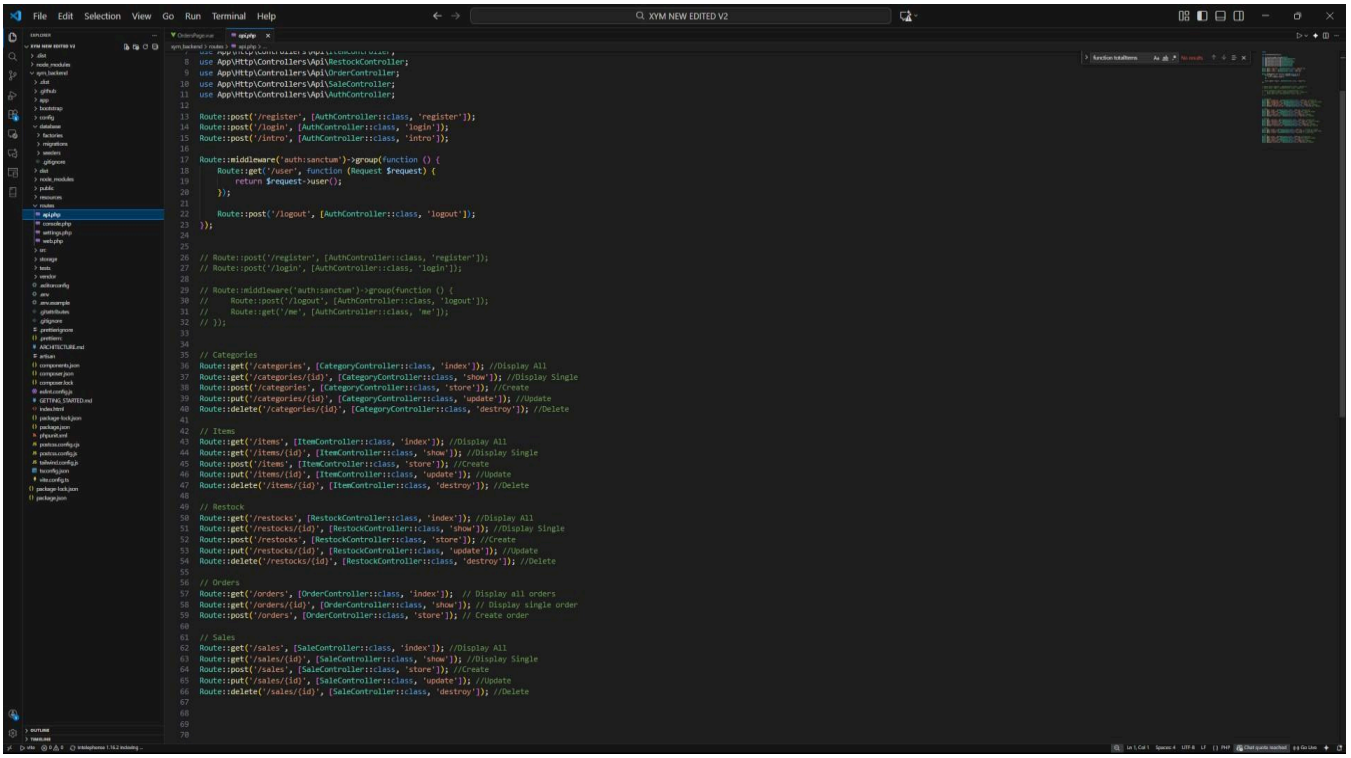


6. I feel the data is secure because it is only accessible inside the store.

1 2 3 4 5



- **Appendix D:** API Documentation



```php
use App\Http\Controllers\Api\RestockController;
use App\Http\Controllers\Api\OrderController;
use App\Http\Controllers\Api\SaleController;
use App\Http\Controllers\Api\AuthController;

Route::post('/register', [AuthController::class, 'register']);
Route::post('/login', [AuthController::class, 'login']);
Route::post('/intro', [AuthController::class, 'intro']);

Route::middleware('auth:sanctum')->group(function () {
    Route::get('/user', function (Request $request) {
        return $request->user();
    });

    Route::post('/logout', [AuthController::class, 'logout']);
});


// Route::post('/register', [AuthController::class, 'register']);
// Route::post('/login', [AuthController::class, 'login']);

// Route::middleware('auth:sanctum')->group(function () {
//     Route::post('/logout', [AuthController::class, 'logout']);
//     Route::get('/me', [AuthController::class, 'me']);
// });


// Categories
Route::get('/categories', [CategoryController::class, 'index']); //Display All
Route::get('/categories/{id}', [CategoryController::class, 'show']); //Display Single
Route::post('/categories', [CategoryController::class, 'store']); //Create
Route::put('/categories/{id}', [CategoryController::class, 'update']); //Update
Route::delete('/categories/{id}', [CategoryController::class, 'destroy']); //Delete

// Items
Route::get('/items', [ItemController::class, 'index']); //Display All
Route::get('/items/{id}', [ItemController::class, 'show']); //Display Single
Route::post('/items', [ItemController::class, 'store']); //Create
Route::put('/items/{id}', [ItemController::class, 'update']); //Update
Route::delete('/items/{id}', [ItemController::class, 'destroy']); //Delete

// Restock
Route::get('/restocks', [RestockController::class, 'index']); //Display All
Route::get('/restocks/{id}', [RestockController::class, 'show']); //Display Single
Route::post('/restocks', [RestockController::class, 'store']); //Create
Route::put('/restocks/{id}', [RestockController::class, 'update']); //Update
Route::delete('/restocks/{id}', [RestockController::class, 'destroy']); //Delete

// Orders
Route::get('/orders', [OrderController::class, 'index']);  // Display all orders
Route::get('/orders/{id}', [OrderController::class, 'show']); // Display single order
Route::post('/orders', [OrderController::class, 'store']); // Create order

// Sales
Route::get('/sales', [SaleController::class, 'index']); //Display All
Route::get('/sales/{id}', [SaleController::class, 'show']); //Display Single
Route::post('/sales', [SaleController::class, 'store']); //Create
Route::put('/sales/{id}', [SaleController::class, 'update']); //Update
Route::delete('/sales/{id}', [SaleController::class, 'destroy']); //Delete
```

- **Appendix E:** Full Design Mockups
  Link: https://tinyurl.com/xymprototy

